INCH-POUND
MIL-P-82691A(OS)
12 December 1989
SUPERSEDING
MIL-P-82691(OS)
25 July 1980
and Amendment-1
21 August 1981

MILITARY SPECIFICATION

PROPELLANT BRAIN, IGNITER

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the propellant grain to be used in the Mark 272 Mod 0 Igniter.

2. APPLICABLE DOCUMENTS

2. 1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto cited in the solicitation (see 6.2).

SPCIFICATIONS
MILITARY

MIL-F-116	Preservation, Methods of
MIL-F-23482	Propellant Grain, Rocket Motor, Mark 30 Mods 1, 2, 3, and 4

Beneficial comments, recommendations, additions, deletions and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730) Indian Head, Maryland 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1340

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-414	Sampling Procedure and Tables for Inspection by Variables for Percent Defective
MIL-STD-453	Inspection ,Radiographic

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS) 5801 Tabor Avenue, Philadephia, FA 19120-5099.)

2.1.2 Other Government documents, drawings l nd publications.Tne following other Government documents drawings and publ ications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation (see 6.2).

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DRAWINGS

Naval Sea Systems Command (Cage Code 10001)

2217185	Propellant Grain, Igniter

(Requests for copies of NAVSEA drawings should be addressed to: Commanding Officer, Naval Ordance Station, Attn: Code 802, Louisville, KY 40214.)

CODE of FEDERAL REGULATIONS

CFR Parts 100-177	Transportation
CFR Parts 178-199	Transportation

(Application for copses should be addressed to the Superintendant of Documents, Government Printing Office, Washington, D. C. 20402)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the

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solicitation Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D412	Method of Tensile Testing of Vulcanized Rubber

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. in the event of l conflict between the text of this document and the references cited herein, the text of this document tales precedence. Nothing in this document, however , superesedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First l rticle. When specified (see 6.2) , a sample shall be furnished for first article inspection and approval (see 4.4 and 6 .3).

3.2 Grain configuration. The grain shall conform with the requirements of the drawings cited herein.

3.3 Chemical composition. The propellant shall have the composition specified in MIL-P-23482.

3.4 Physical properties. The physical properties shall conform to Table I.

Table I. Physical properties requirements.

Property	Minimum	Maximum
Specific gravity	1.690	1.708
Ultimate strain(%)	225	--
Yield stress(psi)	15	--

3.5 Radiographic examination. The igniter propellant grain shall exhibit no evidence of porosity, cracks , fissures or foreign material and shall contain no more than one hole whose largest dimension exceeds 1/16 inch and no more than two holes whose largest dimension exceeds 1/32 inch. There shall be no more than two high density areas whose largest dimension l xceeds 1/16 inch.

3.6 Ballistic properties . The igniter as shown on DL SA 2785411 shall demonstrate the ball istic characteristics shown in Table II when tested at 10 deg. and 120 deg. F.

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TABLE II. Ballistic properties.

Property	10 deg. F		120deg. F	
	Min	Max	Min	Max
Burn time (sec)	.311	---	.244	---
Avg. pressure (psi)	822	---	951	---
Max. pressure (psi)	---	2291	---	2576

3.7 Workmanship. The material shall be uniform in quality and shall be free of voids, cracks, fissures or other defects that could adversely affect its use.

QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or puchase order , the contractor may use his own or any other facilities suitable for the performance of the inspection requiremetns specified herein, unless disapproved by the Gover-nment. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall 1 inspection system or quality program. The absence of any inspection requirements in the specification shall not re leve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply With all requirements of the contract . Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2). all inspection shall be performed under the following conditions:

a. Temperature 18 to 35 deg. C (65 to 95 deg. F)
b. Altitude: Sea level

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c. Vibration: No requirement
d. Humidity: Normal room to 95% max.

4.4 First article inspection. Unless otherwise specified in the contract first article inspection is required and shall be performed at an activity designated by the acquiring activity. First article inspection shall be performed on sample units produced with equipment and procedures normally used in production. The first article sample shall comply with all the requirements specified herein to be accepted. Otherwise the sample shall be rejected (see 3. 1, 4.4.1 and 6.3).

4.4.1 Sample size. The sample shall consist of twenty four igniter propellant grains in accordance with Drawing 2217185.

4.4.2 Inspection routine. Unless otherwise specified, the first article sample shall be subjected to all the quality conformance tests specified herein (see 6.2).

4.5 Quality conformance inspection. Quality conformance inspection for acceptance of material shall consist of all the examinations of 4.6 and the tests of 4.7.

4.5.1 Sampling.

4.5.1.1 Physical properties sampling. Physical properties samples shall be selected in accordance with the contract. (see 6.2 and 6.4). The sampling plan shall be applied to the bullet grain lengths. The selected grains shall be cut to furnish the specimens for the test.

4.5.1.2 Ballistics sampling. Ballistics sampling shall be as specified in the contract (see 6.2 and 6.5).

4.5.2 Inspection lot. The inspection lot shall consist of all 1 grains presented for acceptance at one time produced in a single manufacturing run under homogeneous conditions of manufacture. and made from a single batch of propellant.

4.6 Examinations.

4.6.1 Visual. Measure each grain for compliance with 3.2. Select a sample of grains as specified in the contract and visually determine compliance with 3.7.

4.6.2 Radiographic. The radiographic inspection shall be in accordance with MIL-STD-453 for compliance with the requirements of 3.5.

4.6.3 Preparation for delivery. The packaging shall be examined for conformance to the requirements of 5.

4.7 Test procedures.

4.7.1 Performance properties tests.

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4.7.1.1 Specific gravity test. The specific gravity of the cured propellant shall be determined by the sink-float test Zinc chloride solutions shall be used in this determination in 0. 002 increments. The value of the specific gravity of the cured propellant shall be the specified gravity of the zinc chloride solution in which the sample of propellant just floats.

4.7.1.2 Ultimate strain and yield stress tests. A minimum of two test specimens shall be cut from each propellant sample. 1 e. Each specimen shall be five inches minimum in length and 0.100±0.050 inch in thickness as shown in FIGURE IA. Using Die C of ASTM D412, cut dumb-bells from the specimens as shown in FIGURE IB. Three thickness measurements shall agree within 0.010 inch for acceptable specimens. Reject any specimen that has surface imperfections or 15 damaged. Replace unacceptable samples from the same propellant sample. The testing machine shall have a recorder, adjustable velocity and constant-rate-of-crosshead-movement for performing the ultimate strain and yield stress tests. The crosshead rate shall be O. 10 inch per minute prior to yield and 20 inches per minute from yield to the break point. The average values shall be compared to Table I values for compliance.

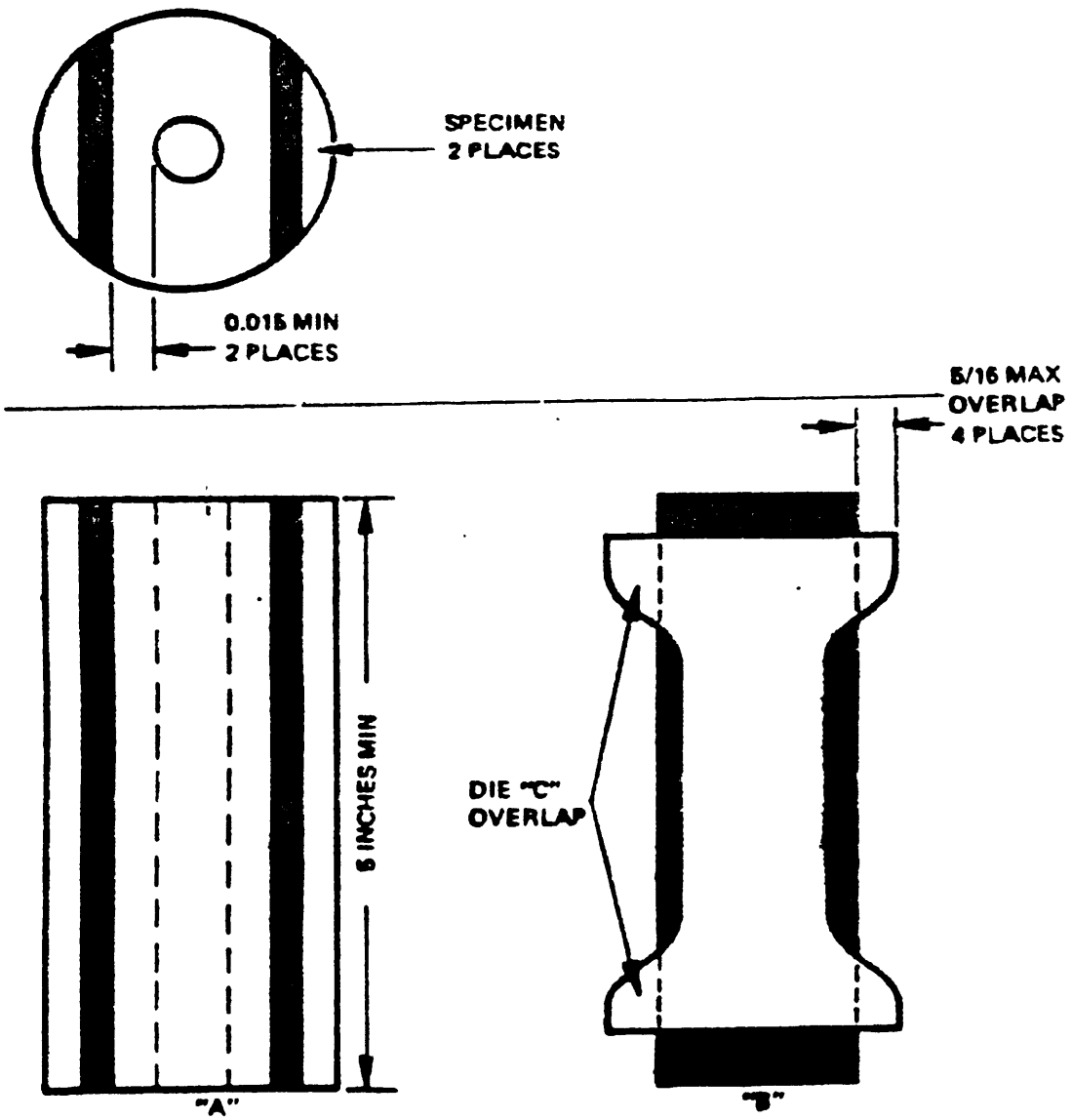


FIGURE 1. Vibration set-up

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4.7.1.3 Ballistic test. The ballistic properties shall be determined when the grain is installed as an igniter assembly for static testing in accordance with drawing SA DL 2785411 less the nozzle assembly. The samples shall be conditioned at 10±5deg. F and 120±5 deg. F as, applicable (see 4.5.1), , for a minimum of one hour prior to testing.

5. PACKAGING

5.1 Packaging and packing. Unless otherwise specified in the contract the packaging and packing shall be in accordance with 49 CFR 100-177 and MIL-P-116, method IIb (see 6.2).

5.2 Harking. Unless otherwise specified in the contract (see 6.2) and in addition to any special marking required by such contract, unit packages and shipping containers shall be marked in accordance With requirements of MIL STD-129 and 49CFR 100-177 and CFR 178-199. Marking shall include, but not be limited to the following:

(a) Manufacturer ‘s name

(b) Material trade name

(c) Gross and net weights

(d) Lot number, serial numbers, and date of manufacture

(e) Number and revision letter of this specification and acceptance drawings

(f) Hazard notice

(g) Unusual storage requirements

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful , but is nor mandatory.)

6,1 Intended use. The propellant covered by this specification is intended for use in the Mark 272 Mod 0 Igniter for the Mark 30 Rocket Motor in accordance with MIL-R-23481.

6.2 Acquisition requirements. Acquisition documents should d specify the following:

a. Title, number, revision letter of this specification

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1 and 2.2).

c. Whether a first article sample is required (see 3.1, 4.4 and 6.3)

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d. Quality acceptance test facility (see 4.1)

e. Inspect Ion conditions (see 4.3)

f. First article test facility and any additional tests (see 4.4 and 4.4.2)

g" Sampling plans for acceptance tests (see 4.5. 1.1, 4.3.1.2, 6.4 and 6.5)

h. Maximum lot size (see 4.5.2)

i. Any special packaging l nd packing (see 5.1)

J. Any special marking (see 5.2)

k.l Safety precautions (see 6.7)

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor 's current inventory (see 3.2), and the number of items to be tested as specified in 4.4. A first article sample is required from each supplier who has not produced the Mk 30 Mod 1, 2, 3 and 4. If igniter grain production has been interrupted for more than 6 months or changes are requested in production methods, a first article sample may be required. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products. who wish to rely on such production or test , must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Physical property sampling. In the past sampling for conformance to the requirements of 3.4 has been based on MIL-STD-105, inspection level S-2, single sampling, normal inspection with an AQL of 1.3 percent (see 4.3.1.1).

6.5 Ballistic properties sampling. In the past, sampling for ballistic testing was performed in accordance with MIL-STD-414 , inspection level 1, normal inspection, single specification limit Table B1, and an AQL of 4.0 percent (see 4.5.1.2).

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6.6 Definitions.

6.6.1 Burn time. The burn time is defined l s the interval between the point where the motor pressure first l ttains 10 percent of its maximum equilibrium value to the tangent point at cutoff.

6.6.2 Average pressure. The average pressure is defined as the integral of the pressure-time curve over the burn time interval divided by the burn time.

6.6.3 Maximum pressure. The maximum pressure is defined l s the highest pressure value attained during the burn time interval.

6.6.4 Batch. A batch is defined l w that quantity of propellant manufactured in one continuous operation at one time under homogeneous conditions by one manufacturer. A batch is equal to or less than a lot.

6.7 Safety precautions. The safety precautions requirements of the “Contractors ‘ Safety Manual for Ammunition, Explosives,and Related Dangerous Material “ (DOD-4145.26M) are l pplicable ‘and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) 1-323.

Note: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore” (OP-5) should be made applicable.

6.8 Subject term (keyword) listing.

Ballistic testing

Environments

Missile round

Sampling

Temperature conditioning

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing Activity
Navy OS
Project No. 1340-N

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